

Energetics

Energy Change in Reaction

Thermal Energy:

“Thermal energy is the energy in an object or system due to the movement of its molecules and atoms.”

Units of Thermal Energy:

Unit of heat or thermal energy used in SI system is the Joule (J).

Another common unit of heat is the calorie.

Calorie:

It is defined as

“the heat or thermal energy required to raise the temperature of one gram of water from 14.5°C to 15.5°C.”

1 Calorie = 4.18 Joules

Thermochemistry

“The branch of chemistry which deals with the heat or thermal energy change- chemical reactions is called thermochemistry.”

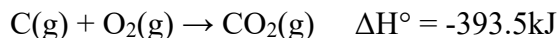
Thermochemical Reactions:

“A chemical reaction which proceeds by the evolution or absorption of heat is called a thermochemical reaction.”

Thermochemical equation

“A balanced chemical equation which also shows heat change of a chemical reaction is called thermochemical equation.”

For example



Types of thermochemical reactions:

There are two types of thermochemical reactions:

(1) Exothermic Reactions

A chemical reaction that proceeds with the evolution of heat is called an exothermic reaction.

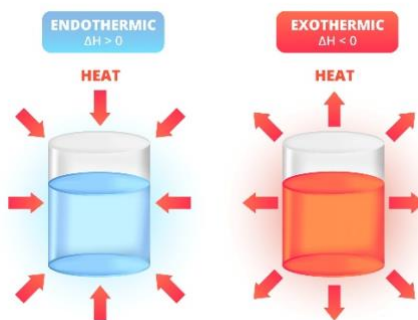
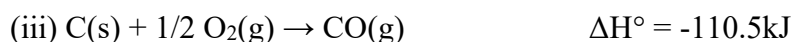
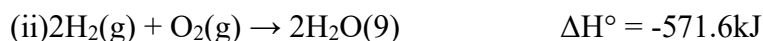
“In an exothermic reaction the chemical system transfers energy to the surroundings as the reactants are converted to products.”

The value of ΔH° is negative for exothermic reaction

Example:

Burning of fuels is a highly exothermic reaction. The energy released can be used to heat a room, or to drive an engine or to cook food.

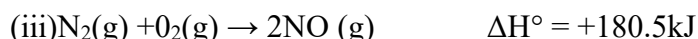
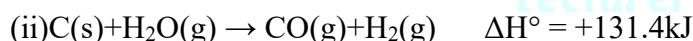
Examples of exothermic reactions are given below:

**(2) Endothermic Reactions:**

“A chemical reaction that proceeds with the absorption of heat is called an endothermic reaction.

In these reactions heat is transferred from surrounding to the system.”

The value of ΔH° is positive for endothermic reaction.

Examples:**Heat of Reaction:**

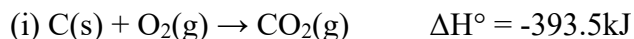
“The amount of heat evolved or absorbed in a chemical reaction, when molar quantities of reactants and products are same in a chemical reaction is called heat of reaction.”

The amount of energy returns, absorbed depends on the amounts of substances that react. For example, as more fuel is burned more energy is released.

Energy changes are expressed in terms of molar quantities of reactants and products.

Standard enthalpy change

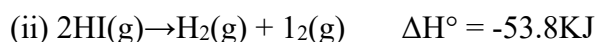
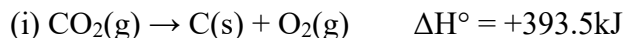
“Heat of reaction measured at 25°C (or 298K) and one atmospheric pressure is known as standard enthalpy change. It is denoted by ΔH° .”



If a reaction is exothermic when going in one direction, it will be endothermic in the reverse direction.

When a reaction is reversed, the magnitude of ΔH° remains the same but its sign changes.

Examples:



Conditions for Standard Heat of Reaction:

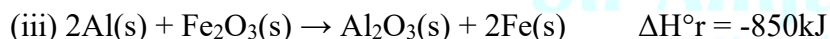
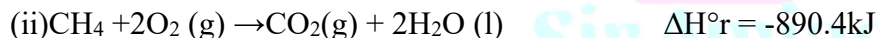
1. Standard state for a gas is 1 atm.
2. Standard state for an element or a compound is the most stable physical state at 1 atm. and 25°C (298 K).
3. Standard state for a substance in aqueous solution is 1M concentration.

1. Standard Enthalpy of Reaction (ΔH°_r):

“The enthalpy changes in a chemical reaction, when reactants and products are in their standard states and their molar quantities are same as shown by the balanced chemical equation is called standard enthalpy of reaction.”

Examples:

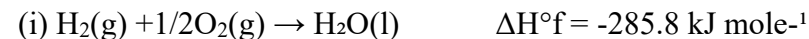
Examples are given below:

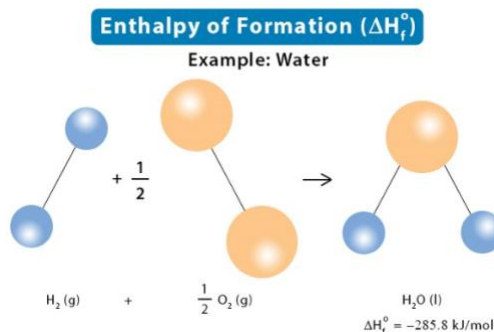


2. Standard Enthalpy of Formation ΔH°_f :

It is defined as

“The enthalpy change that involve the formation of one mole of a compound from its elements with all substances in their standard states is standard enthalpy of formation.”





3. Standard Enthalpy of Combustion (ΔH°_c)

“The enthalpy changes when one mole of a substance is completely burnt in excess of oxygen under standard conditions is called standard enthalpy of combustion.”

Writing thermochemical equation for enthalpy of combustion

- ✓ Write 1 mole of the element or compound and oxygen as reactant.
- ✓ Write oxides of the given element or oxides of elements present in the compound as products.
- ✓ Show standard states of all the substances.
- ✓ Finally, balance the atoms.

Example:

- (i) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ $\Delta H = 393.5 \text{ kJ mole}^{-1}$
 (ii) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ $\Delta H = -890.4 \text{ kJ mole}^{-1}$
 (iii) $\text{H}_2(\text{g}) + 1/2\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ $\Delta H = -285.8 \text{ kJ mole}^{-1}$

4. Standard Enthalpy of Atomization ($\Delta H^\circ_{\text{at}}$)

“The enthalpy changes when one mole of gaseous atoms are formed from its element under standard conditions is called standard enthalpy of atomization.”

Example:

- (i) $1/2\text{H}_2(\text{g}) \rightarrow \text{H}(\text{g})$ $\Delta H^\circ_{\text{at}} = +218 \text{ kJ mole}^{-1}$
 (ii) $1/2 \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ $\Delta H^\circ_{\text{at}} = +121 \text{ kJ mole}^{-1}$

5. Standard Enthalpy of Neutralization (ΔH°_n)

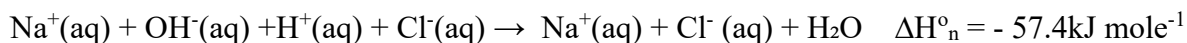
It is defined as,

The amount of heat evolved when one mole of H^+ ions from an acid combine with one mole of OH^- ions from a base to form one mole of water under standard conditions.

Example:



Strong acids and bases ionize completely in their aqueous solutions. The only change which occurs is the formation of water. Na and Cl⁻ ions remain in solution.



Or



7. Standard Enthalpy of first Electron Affinity

“The energy released when one mole of gaseous atoms gains an electron to form one mole of gaseous ions with a single negative charge, under standard conditions.”

For example;



8. Lattice Energy

“The energy released when one mole of an ionic compound(solid) is formed from its constituent gaseous ions is called lattice energy.”

Example



Bond Energy

Bond Dissociation Energy:

“Energy required when one mole of particular bond breaks apart to form a neutral atom is called bond dissociation energy.”

Bond energy:

“Energy released when one mole of neutral atom form bond is called bond energy.”

Enthalpy Change:

“The sum of the bond dissociation energies of reactant minus the sum of the bond energies of the products is called enthalpy change.”

$$\Delta H = \Sigma H (\text{bonds broken}) - \Sigma H (\text{bonds formed})$$

Example 6.3

We can theoretically calculate bond energy of a bond from the known bond energy data. For example, we can calculate bond energy on H-Cl as follows;

Bond dissociation energy of H₂ = 436 kJ/mol.

Bond dissociation energy of Cl₂ = 243 kJ/mol.

Bond energy of HCl = ?

Solution

Total energy absorbed in bond breaking = Total energy released in bond formation
 Bond dissociation energy of H-H + Bond dissociation energy of Cl-Cl = 2(Bond energy of H-Cl)

$$436 + 242 = 2(\text{Bond energy of H-Cl})$$

$$679 = 2(\text{Bond energy of H-Cl})$$

$$\text{Bond energy of H-Cl} = 339 \text{ kJ/mol.}$$

Variability in Bond Energies

- ✓ Some bond energies can vary depending on the specific molecules involved and the environment in which the bond occurs.
- ✓ Diatomic molecules such as H₂, O₂, N₂ and Cl₂, binding energies can be considered relatively accurate. This is because these molecules are made up of identical atoms and the bond dissociation energy is the same for every bond in the molecule.
- ✓ But in complex molecules or molecules in different environments, binding energies can be more approximate. This is because the bond strength can be affected by factors such as neighboring atoms, molecular geometry and inductive effect of functional groups.
- ✓ For example, the C-C bond energy in ethane (CH₃CH₃) can be considered approximate because it
- ✓ depends on the specific isomer (e.g., linear or branched) and the general structure of the
- ✓ molecule.
- ✓ Similarly, the binding energy of a CH bond can vary depending on the chemical
- ✓ environment.
- ✓ The O-H bond is present in various compounds. The bond energy of an OH bond in methanol (CH₃OH) is different from that of H₂O due to difference in molecular structure and chemical environment.

Hess's Law**Statement:**

"It states that the enthalpy change in a chemical reaction is same whether the reaction takes place in a single step or in several steps."

Mathematically

$$\Sigma \Delta H (\text{Cycle}) = 0$$

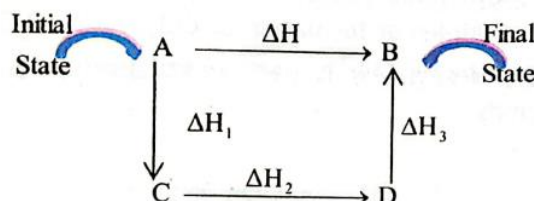
G.H. Hess stated this law in 1840.

Explanation:

Enthalpy is a state function; therefore, enthalpy change in a chemical reaction in going from some initial state to some final state is independent of the path followed by the reaction. Thus, in going from a particular set of reactants to a particular set of products, the change in enthalpy is the same whether the reaction takes place in one step or in many steps. This principle is known as Hess's law.

Suppose a reactant A changes into the product B in one step and enthalpy change in this step is ΔH .

Now suppose this change takes place in three steps, involving a change from A to C, C to D, and finally D to B as shown below:



If ΔH_1 , ΔH_2 and ΔH_3 are enthalpy changes in these steps, as shown in the figure.

Then according to the Hess's Law.

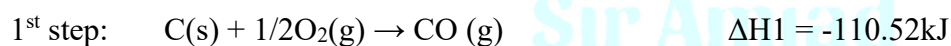
$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

Example 6.6

Combustion of C to CO_2 evolves 393.5 KJ of energy.



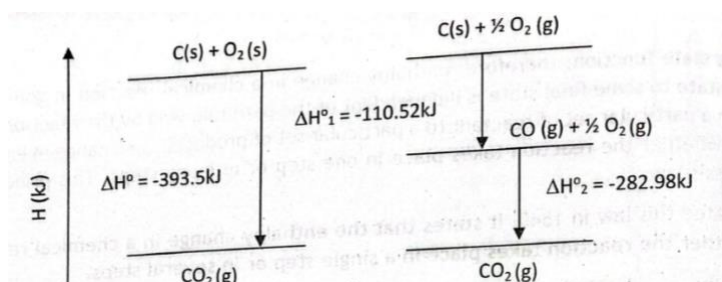
This reaction may take place in two steps.



Net reaction



Thus, the enthalpy change in a chemical reaction is independent of the path followed.



Born Haber Cycle

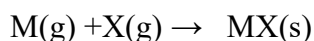
It can be defined as,

“Series of hypothetical steps and their enthalpy changes needed for the formation of ionic compounds from its constituents’ elements is called Born Haber cycle.”

This is special application of Hess's law to binary ionic compounds. It helps us to calculate lattice energies of binary ionic compounds (M'X').

Lattice energy:

“The change in energy that takes place when gaseous ions are combine together to form one mole of an ionic solid is called lattice energy.”



Born Haber cycle for NaCl

Step-1: Sublimation of solid sodium

The energy of sublimation for Na(s) is 108 kJ/mole



Step-II: Ionization of Na(g) atom to form Na (g) ion

This process corresponds to the first ionization energy for Na.



Step-III: Dissociation of Cl₂ molecules

We need to form one mole of Cl atoms by breaking the Cl-Cl bond in 1/2 mole of Cl₂ molecules. The energy required to break this bond is 121 kJ/mole and is known as enthalpy of atomization for Cl₂.



All these three steps are endothermic and are drawn upward in the figure

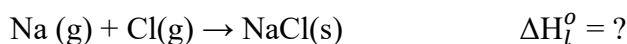
Step-IV: Formation of Cl(g) ion

Energy is released in this step equal to the electron affinity for Cl.



Step-V: Formation of solid NaCl

This corresponds to the lattice energy (ΔH) for NaCl(s) which is to be calculated.



Overall energy change

Sum of these five steps gives the overall reaction and the sum of the individual energy changes give the overall energy change.

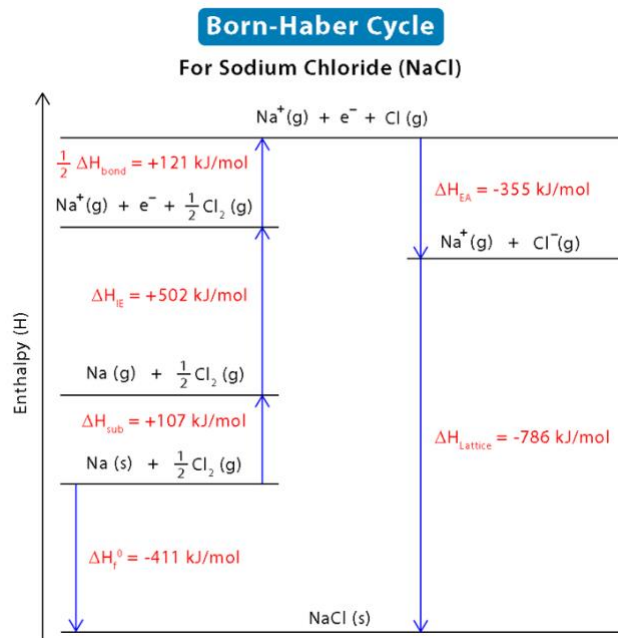
$$\Delta H_f^{\circ} = \Delta H_s^{\circ} + \Delta H_{I.E}^{\circ} + \Delta H_{at}^{\circ} + \Delta H_{E.A}^{\circ} + \Delta H_l^{\circ}$$

$$-411 \text{ kJ} = +108 \text{ kJ} + 496 \text{ kJ} + 121 \text{ kJ} + (-349 \text{ kJ}) + \Delta H_l^{\circ}$$

$$\Delta H_f^\circ = -411 \text{ kJ} - (108 \text{ kJ} + 496 \text{ kJ} + 121 \text{ kJ} - 349 \text{ kJ})$$

$$\Delta H_f^\circ = -787 \text{ kJ/mole}$$

Thus, lattice energy of NaCl is 787 kJ/mole.



Factors Affecting Lattice Energy

- ✓ The magnitude of lattice energy is influenced by the charge density of an ion.
The charge density is defined as the ratio of ionic charge to the ionic radius of an ion.

$$\text{Charge density} = \frac{\text{Ionic charge}}{\text{ionic radius}}$$

Ionic Charge

- ✓ As the magnitude of the ionic charge increases, the lattice energy also increases.

$$\text{Ionic Charge} \propto \text{Lattice energy}$$

- ✓ The reason is that the higher charges have strong electrostatic attractions, and they release higher energy when they come together to form an ionic lattice.

For example

MgO has a higher lattice than NaCl. This is because Mg⁺² and O⁻² ions have a larger magnitude of charge as compared to the Na⁺¹ and Cl⁻¹ ions.

Ionic radius

- ✓ Smaller ions have a higher charge density, exert a stronger attraction on water molecules.

$$\text{Ionic radius} \propto \frac{1}{\text{Enthalpy of hydration}}$$

- ✓ Larger ions spread out the charge distribution and exert a weaker attractive force on water molecules. Thus, less energy is released when they hydrate.
- ✓ For example, the hydration enthalpy of MgSO_4 is greater than that of CaSO_4 . Sulphate ions are common in these compounds, their hydration enthalpy difference must be due to Mg^{2+} and Ca^{2+} ions. Both Mg and Ca belong to group 2 of the periodic table. Mg is in the group below Ca, so the Mg^{2+} ion is smaller than the Ca^{2+} ion.
- ✓ Thus, the attraction of water molecules to Mg^{2+} ions is stronger than to Ca^{2+} ions. As a result, the hydration enthalpy of MgSO_4 is more exothermic than that of CaSO_4 .

Role of Hydration in the Dissolving Process

Enthalpy of hydration:

“The amount of heat released when one mole of a gaseous ion dissolves in a large amount of water to form an infinitely dilute solution.”

Enthalpy of dissolution:

“The amount of heat that is released or absorbed during the dissolving process at constant pressure is called enthalpy of dissolution.”

- ✓ The energy change of solution formation depends on three types of interactions between solute-solute, solvent-solvent and solute-solvent. When solution is formed the interaction between solute particles are broken.
- ✓ When solute is dissolved in solvent the interaction between solvent molecules are broken and energy is required to break the solute interaction so it is an endothermic process.
- ✓ At the same time interaction between the solute and solvent occurs and energy is released. So, it is an exothermic process.
- ✓ Both these processes determine whether the dissolution process is endothermic or exothermic.
- ✓ Therefore, when a solute dissolves in water, the attractive force of the solute particles must be overcome to break the solute particles out of the solid lattice.
- ✓ A strong force must bind these separate solute particles to the water molecules in the solution.
- ✓ Therefore, the heat of solution measures the net flow of energy that occurs when a substance dissolves.
- ✓ The energy required to break the solute particles is equal to the lattice energy ($\Delta H_{\text{lattice}}$) of the solute.
- ✓ On the other hand, the energy released when solute particles bind to water molecules in solution is equal to the enthalpy of hydration (H_{hyd}).

- ✓ So, enthalpy of solution $\Delta H_{\text{solution}}$ is the difference in $\Delta H_{\text{lattice}}$ and ΔH_{hyd} .
- ✓ $\Delta H_{\text{solution}} = \Delta H_{\text{lattice}} - \Delta H_{\text{hyd}}$
- ✓ When $\Delta H_{\text{solution}}$ is very large, the solute is unlikely to be soluble.
- ✓ But if $\Delta H_{\text{solution}}$ is moderately positive, then solute dissolves.
- ✓ **For example**
- ✓ When ammonium nitrate dissolves in water 25.7 kJ/mole energy is absorbed and the flask becomes cold. This is because crystal binds NH_4^+ and NO_3^- ions more tightly than the solution.
- ✓ $\text{NH}_4\text{NO}_3(\text{s}) + \text{XH}_2\text{O}(\text{l}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq}) \quad \Delta H_{\text{solution}} = 25.7 \text{ kJ/mole}$
- ✓ **Exothermic process.**
- ✓ **For example**, when NaOH dissolves in water 44.5 kJ/mole energy is released and the flask becomes hot.
- ✓ $\text{NaOH}(\text{s}) + \text{XH}_2\text{O}(\text{l}) \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq}) \quad \Delta H_{\text{soln}} = -44.5 \text{ kJ/mole}$

Factors Affecting Enthalpy of Hydration:

The charge density is defined as the ratio ionic charge to the ionic radius of an ion.

Charge density = Ionic charge / ionic radius

Therefore, two factors affect enthalpy of hydration.

1. Ionic charge

Ions with a higher charge exert stronger attractive forces on water molecules through ion-dipole interactions, resulting in a higher enthalpy of hydration.

Ionic Charge $\propto$ Enthalpy of hydration

For example, the hydration enthalpy of MgO is more exothermic than that of NaCl. These compounds are ionic compounds and consist of Mg^{2+} , O^{2-} ; and Na^+ , Cl^- ions, respectively. Both Mg^{2+} and O^{2-} have a greater charge than Na^+ and Cl^- so the hydration enthalpy of MgO is more exothermic than NaCl.

2. Ionic radius

Smaller ions have a higher charge density, exert a stronger attraction on water molecules.

Ionic radius $\propto \frac{1}{\text{Enthalpy of hydration}}$

Larger ions spread out the charge distribution and exert a weaker attractive force on water molecules.

Thus, less energy is released when they hydrate.

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Thus, the attraction of water molecules to Mg^{2+} ions is stronger than to Ca^{2+} ions. As a result, the hydration enthalpy of MgSO_4 is more exothermic than that of CaSO_4 .

Entropy

"The degree of disorder and randomness of a system is called its entropy."

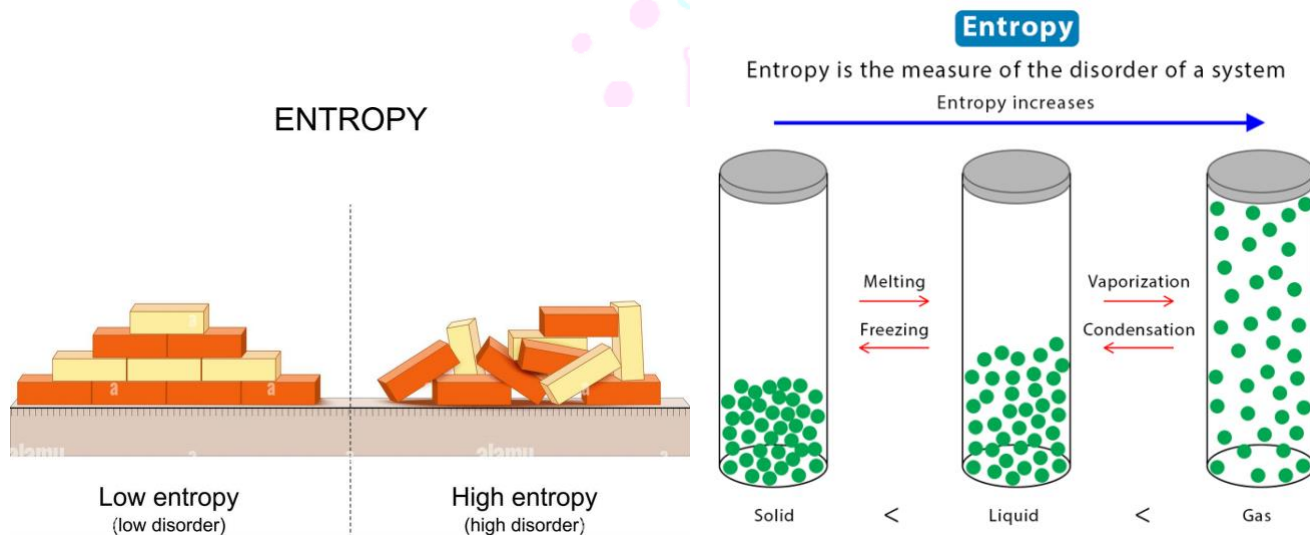
It is marked with the symbol "S".

As a system moves from a more ordered state to a less ordered state, the entropy of the system tends to increase.

Example:

In ice the particles are fixed and well-ordered so they have the minimum entropy. But when the ice is heated the particles get energy and gain more freedom which causes the increase in the disorderness.

The entropy change of the system is denoted by ΔS .



- ✓ If the entropy of the system increases, this is represented as a positive change $\Delta S > 0$
- ✓ When the entropy of a system decreases, it is represented as a negative change $\Delta S < 0$
- ✓ In general, when the temperature of a system increases, its entropy also increases, its $\Delta S > 0$.
- ✓ As the temperature of the system decreases, its $\Delta S < 0$.
- ✓ The value of ΔS is positive when number of molecules of product are greater than reactants.
- ✓ The value of ΔS is negative when number of molecules of reactants are less than products.

For More Information:

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